

## BREATHEBOX · DESIGN DECISION RECORD

# BreatheBox recommendations accepted

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## DOCUMENT CONTROL

|             |                                                           |         |                |
|-------------|-----------------------------------------------------------|---------|----------------|
| Document ID | BBX-DDR-002                                               | Version | 0.2            |
| Status      | <b>Draft</b>                                              | Date    | 2026-09-26     |
| Author      | Amish Chadha                                              | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept                             |         |                |
| Repository  | github.com/BoujeeEnjinia1701/breathebox                   |         |                |
| Source      | docs/decisions/0002-recommendations-accepted.md @ d61ee78 |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                        |
|---------|------------|--------------|---------------------------------------------------------------------------------------------------------------|
| 0.2     | 2026-09-26 | Amish Chadha | Budget set to \$255 to cover the priced BOM: decided by Amish, 2026-09-26 (O3 closed)                         |
| 0.1     | 2026-09-25 | Amish Chadha | Record the recommendations accepted by Amish on 2026-09-25, what changed in the repo and the items still open |

## 0002: Recommendations accepted

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- **Date:** 2026-09-25
- **Status:** accepted. Amish accepted every recommendation in `docs/REVIEW.md` and BBX-DDR-001. Items without a recommendation stay proposed, awaiting Amish, except the budget gap (O3), decided by Amish on 2026-09-26.

### ■ Context

On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Every item in `docs/REVIEW.md` and BBX-DDR-001 that was awaiting Amish or adopted for TRL 3 pending his review, and that carried a recommendation, is now decided: go with recommendation. Where a recommendation offered several options, the recommended option is the decision. Items with no recommendation are not decided here. TRL 4 remains on hold by Amish's instruction, so decided items that need building, testing or purchasing are recorded as decided but on hold.

### ■ Decision

*Table 1. Items decided by Amish, 2026-09-25: go with recommendation.*

| #  | Item (source)                                                                  | Decision                                                                                                                                                                          | What changed in the repo                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Budget (TRL 2 item 1; DDR-001 A1)                                              | Raise <code>budget_usd</code> from \$220 to \$250; keep the bought core and the CO2 sensor                                                                                        | <code>project.yaml</code> <code>budget_usd</code> 220 to 250; README budget line; BBX-REQ-001 R15 target \$250; BBX-CAL-001 section 9 and <code>results.csv</code> ; BBX-PRB-001 constraints; <code>bom/bom-notes.md</code> ; blueprint key figures. R15 is still not met (\$255, \$5 over)                                                                                                                                                                                                   |
| 2  | Window type (TRL 2 item 2; A2)                                                 | Vertical sliding sash first; casement and tilt-and-turn later                                                                                                                     | Wording only (BBX-PRB-001, BBX-REQ-001 R8, BBX-PRC-001); design already sash-first                                                                                                                                                                                                                                                                                                                                                                                                            |
| 3  | Core type (TRL 2 item 3; A3)                                                   | Bought polymer counterflow plate core                                                                                                                                             | Wording only (BBX-PRC-001); already in BOM line 2                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4  | Frost strategy (TRL 2 item 4; A4)                                              | Slow the supply fan, with the combustion-appliance warning                                                                                                                        | Wording only; already in BBX-CAL-001 section 6 and the safety notes                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5  | Nominal airflow (TRL 2 item 5; A5)                                             | 50 m <sup>3</sup> /h with CO2-driven boost to 70 m <sup>3</sup> /h                                                                                                                | Wording only; already the design case                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 6  | Controller (TRL 2 item 6; A6)                                                  | ESP32-C3 class module with an SCD41 class sensor, checked on CalRig                                                                                                               | Wording only. The CalRig sensor check is TRL 4 work: decided, on hold                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 7  | Noise, R6 (TRL 3 item 2; DDR-001 O2)                                           | Option (a) quiet night mode near 30 to 32 m <sup>3</sup> /h, plus option (b) larger, slower fans or a lined silencer checked against a real fan datasheet before any change to R6 | Firmware rule added: night mode holds both streams at 32 m <sup>3</sup> /h and suppresses the CO2 boost. BBX-CAL-001 v0.2 ( <code>sizing.py</code> ): 30.1 dB(A) clean, 32.0 dB(A) loaded; CO2 1,295 ppm overnight, 712 ppm 24 h mean. BBX-PRC-001, BBX-REQ-001 R4 and R6 status, BBX-DWG-001 note (Rev P1 to P2), blueprint key figures, README. R6 target unchanged and still not met at 50 m <sup>3</sup> /h. Option (b) needs a chosen fan and its datasheet: decided, on hold with TRL 4 |
| 8  | Larger blowers, about 100 m <sup>3</sup> /h free air and 400 Pa (TRL 3 item 4) | Keep                                                                                                                                                                              | Already in BOM lines 3 and 4 and BBX-CAL-001; wording updated in BBX-PRC-001                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 9  | 2 A time-delay input fuse (TRL 3 item 4)                                       | Keep                                                                                                                                                                              | Already in BOM line 13; wording updated in BBX-PRC-001                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 10 | Firmware fan speed cap at the 80 m <sup>3</sup> /h need (TRL 3 item 4)         | Adopt as a firmware rule                                                                                                                                                          | BBX-PRC-001 (How it works, step 6) and BBX-CAL-001 section 4 now state it as a rule. No firmware is written (TRL 4, on hold)                                                                                                                                                                                                                                                                                                                                                                  |
| 11 | Install restriction for rooms whose door seals airtight (TRL 3 item 4)         | Keep                                                                                                                                                                              | Already in the safety notes of BBX-PRC-001 and BBX-CAL-001                                                                                                                                                                                                                                                                                                                                                                                                                                    |

No recommendation changed the pitch or problem wording, so `project.yaml` and `README.md` keep them.

## Items still open

Table 2. Items still proposed, awaiting Amish (no recommendation was made).

| #  | Item                                                                                                                                                                           | Status                                                        |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| O1 | First co-design partner and region (for example a tenants' group or social landlord in the UK or Canada)                                                                       | Proposed, awaiting Amish                                      |
| O3 | Budget gap: priced BOM \$255 against the new \$250 (accept about \$255, drop to smaller blowers and lose the 70 m³/h boost, or find \$5 elsewhere)                             | Decided by Amish, 2026-09-26: budget set to \$255 (see below) |
| O4 | Fan position: both fans in the room-end plenum (exhaust side of the core at higher pressure, so any core leakage reaches the supply) versus the exhaust fan at the outdoor end | Proposed, awaiting Amish. No change made                      |

## Budget approved, 2026-09-26

On 2026-09-26 Amish wrote: "i approve all the budget items."

- Budget set to \$255 to cover the priced BOM: decided by Amish, 2026-09-26. This closes O3. `project.yaml` `budget_usd` 250 to 255; BBX-REQ-001 R15 target \$255, status Not met to Met; BBX-CAL-001 v0.3 (`sizing.py` and `results.csv` rerun); BBX-PRB-001, BBX-PRC-001, `README`, `bom/bom-notes.md` and the blueprint key figures updated. The BOM and geometry are unchanged.

## ■ Consequences

- Requirement status (BBX-CAL-001 v0.3): met 11, not met 1 (R6 at 50 m³/h), not verifiable at TRL 3 3 (R9, R13, R14). The night mode narrows the R6 gap in use but not as R6 is written. R15 is met since the 2026-09-26 budget approval.
- The geometry and BOM are unchanged. The GA drawing moves to Rev P2 for the night mode note only; STEP and STL were re-exported from the unchanged model.
- Cross-repo: the SCD41 class sensor check on CalRig is decided and on hold with TRL 4. No other repo is affected, and none was edited.
- `trl: 3` and `trl_target: 3` are unchanged.